
Phase-Conditioned Student-t Mixture Regression for X-Trans Demosaicing

A robust joint spatial-chromatic patch prior with
exact CFA data consistency

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The Fujifilm X-Trans color filter array samples red, green, and blue on an irregular 6 by 6 lattice. Its larger fundamental cell complicates both hand-designed interpolation and the direct transfer of Bayer demosaicers. This paper presents a phase-conditioned Student-t Gaussian-mixture-regression (TGMR) demosaicer that predicts the two missing center colors directly from the 49 physically observed samples in a 7 by 7 neighborhood. Eighteen phase-specific models are trained on synthetic X-Trans observations of linearized natural-image patches. Heavy-tailed component likelihoods replace the brittle Gaussian responsibilities of an earlier mixture regressor, while the conditional predictor remains affine within each component. A validation-selected K32/S9/q8 reduction first scores 32 components on a central 3 by 3 marginal, then evaluates eight full 49-dimensional experts. Exact algebraic specialization for degrees of freedom 3 and posterior temperature 4 enables an AVX2/FMA and OpenMP implementation without approximating the reviewed statistical model. On fixed center coordinates from 20 held-out BSDS sources, the reduced method reaches 35.056 dB, 3.002 dB above a global joint spatial-chromatic LMMSE baseline, while preserving the physically measured center sample exactly. It remains robust on frozen chromatic, star-field, and analytical controls, although a tiny-star control remains 0.271 dB below the LMMSE baseline. A 39.96-megapixel standalone mosaic is processed in 5.83 s on a Ryzen 9 5900X; integration into RawTherapee processes a 40.14-megapixel RAF in 11.28 s end to end, approximately three times the Markesteyn baseline. The results support Student-t mixture regression as a useful, CPU-viable experimental X-Trans demosaicer, while leaving model redistribution and cross-camera validation unresolved.

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1 Introduction

A single-sensor digital camera measures only one spectral component at each pixel. Demosaicing reconstructs the two missing components from neighboring measurements and from assumptions about natural-image structure. The problem is ill-posed: multiple full-color images can produce the same mosaic, and high-frequency spatial content can overlap the chromatic aliases introduced by the color filter array (CFA).

Most published demosaicing work assumes a 2 by 2 Bayer lattice. Fujifilm’s X-Trans CFA instead repeats over a 6 by 6 cell and places green more densely and less regularly. Rafiazari and Dubois analyzed the corresponding spectral replicas and proposed dedicated X-Trans interpolation filters [1]. RawTherapee’s principal classical X-Trans implementation is attributed in its source to Frank Markesteijn; no separate formal Markesteijn paper was identified, so this work cites the implementation rather than inventing a publication [2]. Both approaches can produce convincing photographs, but difficult periodic textures, thin colored detail, and metallic edges remain challenging.

Population statistics offer another route. Portilla, Otaduy, and Dorronsoro showed that joint spatial-chromatic covariance learned from complete images can produce compact linear CFA predictors [3]. Sandeep and Jacob extended the idea to a joint-color Gaussian mixture model (JCS-GMM), choosing a local Gaussian from observed CFA values and applying its conditional mean [4]. A mixture is attractive because a single covariance must average incompatible structures such as smooth regions, oriented edges, and chromatic texture.

Yet an observed-sample Gaussian likelihood can become confidently wrong when a patch is atypical or when CFA masking removes the evidence separating two components.

This paper develops a robust mixture-regression method specifically for X-Trans. Each of the 18 distinct phase/orientation observation contracts has its own low-dimensional joint model over 49 observed CFA samples and two missing center colors. The Gaussian mixture is continued into a multivariate Student-t mixture using fixed-degree-of-freedom expectation conditional maximization (ECM). Student-t tails soften component responsibilities for unusual patches without changing the affine conditional location within an expert. Model reduction then combines a genuinely retrained 32-component mixture with observable coarse-to-fine component shortlisting.

The documented contributions are:

1. **Phase-conditioned task reduction.** Rather than learning a full 147-dimensional RGB-patch density, the method trains 18 joint regressors over exactly the physically observed 7 by 7 CFA values and the two missing center targets. No initial demosaic, Bayer conversion, or target-RGB leakage is used.
2. **Robust Student-t responsibilities for X-Trans regression.** Fixed Student-t tails repair a held-out external failure of the Gaussian model while increasing population quality and lowering tail error.
3. **Observable coarse-to-fine inference.** A central S9 Student-t marginal selects eight candidates from a genuinely trained $K=32$ model; the final posterior is evaluated on the complete 49-sample observation.
4. **Exact CPU specialization.** For the frozen degrees of freedom and posterior temperature, shortlist ranking and posterior weights are rewritten without per-pixel logarithms or exponentials. AVX2/FMA, same-component bucketing, OpenMP, and bounded streaming yield practical CPU throughput without changing the reviewed model.
5. **End-to-end open-source integration.** The model is authenticated, phase-matched against the camera’s actual CFA, evaluated on RawTherapee’s camera-linear sensor scale, and backed by complete Markestijn fallback [5].

These are implementation- and experiment-bounded contributions. The paper does not claim that the constituent statistical ideas are new in isolation, that the combination is the first possible one, or that the work establishes patent novelty.

2 Related Work

2.1 Classical and X-Trans demosaicing

Classical demosaicers exploit smooth color differences, directional interpolation, residual correction, frequency separation, or learned linear statistics. Residual interpolation predicts a guide image and interpolates the residual between observed and guided samples; Kiku et al. demonstrated that this view can outperform direct color-difference interpolation on Bayer data [6]. Portilla et al. instead learned the joint covariance of neighboring colors and derived one local LMMSE predictor for each CFA phase [3]. This learned-linear formulation is especially relevant here: the $K=1$ phase-conditioned control is the affine Gaussian/LMMSE limit of the proposed mixture model.

X-Trans requires explicit treatment of its 6 by 6 periodicity. The frequency-domain analysis of Rafinazari and Dubois identifies overlapping luminance and chrominance replicas that are not present in the same form for Bayer [1]. Markesteijn’s RawTherapee implementation uses directional interpolation and multiple passes over candidate reconstructions [2]. The proposed method does not modify either algorithm; they serve as classical comparisons and, in the integration, Markesteijn remains the complete fallback.

2.2 Joint-color Gaussian patch priors

JCS-GMM learns full-covariance Gaussian components over concatenated RGB patches and conditions a selected component on the observed Bayer samples [4]. Its published configuration uses 150 components, 6 by 6 patches, hard observed-CFA component selection, overlapping aggregation, and image-adaptive updates. Our earlier X-Trans reproduction deliberately isolated the local probabilistic core. A soft posterior mean improved upon hard MAP, but a full 147-dimensional RGB-patch mixture generalized poorly: likelihood optimization modeled many variables that were irrelevant to the two missing center components.

Zoran and Weiss use Gaussian patch mixtures in expected patch log likelihood (EPLL), coupling a patch prior to a whole-image inverse problem [7]. The present work does not perform EPLL or iterative whole-image refinement. It asks a narrower question: can a fixed learned mixture predict a center RGB value directly and safely from an X-Trans neighborhood? This separation prevents overlap iteration or image-specific adaptation from hiding weaknesses in the local prior.

2.3 Heavy-tailed mixtures and learned demosaicing

Peel and McLachlan formulate robust mixture modeling with multivariate Student-t components and latent scale weights [8]. Ding gives the conditional multivariate Student-t distribution used here to separate the conditional location from its observation-dependent predictive scale [9]. We apply these established results to phase-conditioned X-Trans mixture regression and examine the effect of heavy-tailed responsibilities on demosaicing safety.

Deep networks learn far richer nonlinear priors. Gharbi et al. include an X-Trans-specific network and emphasize mining difficult training patches [10]. Kokkinos and Lefkimmiatis derive an optimization-inspired iterative residual network that can accommodate non-Bayer patterns [11]. Such systems can achieve excellent output, but they require licensed weights and a neural runtime. TGMR instead evaluates a compact fixed statistical model with ordinary C++11, OpenMP, and optional AVX2/FMA. The comparison is architectural rather than a claim that a mixture model generally outperforms modern neural demosaicers.

3 X-Trans Observation Model

Let a complete linear-light RGB image be

$$x : \Omega \rightarrow \mathbb{R}^3,$$

and let $c(r) \in \{R, G, B\}$ be the color measured at pixel r by a 6 by 6 X-Trans CFA. The scalar mosaic is

$$y(r) = e_{c(r)}^\top x(r),$$

where e_c selects one color component. At a target pixel r_0 , a 7 by 7 window contains one physical observation at every spatial location. Vectorize those 49 values in row-major spatial order as $y_p \in \mathbb{R}^{49}$, where p denotes the local X-Trans phase pattern. The target vector $t_p \in \mathbb{R}^2$ contains only the two unmeasured center components. Its channel order is fixed by the measured center color:

Center sample	Target order
R	G, B
G	R, B
B	R, G

Across translated and transformed X-Trans representations there are 18 unique 7 by 7 observation patterns in the experiment. A model is trained separately for each pattern. At runtime the complete local CFA pattern, not merely the center color, selects the phase bank. Thus equally colored center samples with different surrounding lattices do not share a regressor.

3.1 Observable common DC

Absolute patch brightness consumes mixture capacity and makes likelihoods sensitive to global intensity. For each phase, let $I_{p,c}$ be the indices of the observed values whose physical CFA color is c . The common observable DC is

$$d(y_p) = \frac{1}{3} \sum_{c \in \{R, G, B\}} \frac{1}{|I_{p,c}|} \sum_{i \in I_{p,c}} y_{p,i}.$$

The scalar d is subtracted from all 49 observations and from both target components during training. It is added back after prediction. Only measured CFA values enter this operation. This formulation was selected on validation and was stronger than either an absolute-RGB model or subtracting the mean of all observations without color balancing.

4 Phase-Conditioned Student-t Mixture Regression

For one phase, define the task-reduced vector

$$z = \begin{bmatrix} y \\ t \end{bmatrix} \in \mathbb{R}^{51}.$$

The prior is a K -component multivariate Student-t mixture

$$p(z) = \sum_{k=1}^K \pi_k \text{St}_\nu(z; \mu_k, \Sigma_k),$$

with fixed degrees of freedom $\nu = 3$. Partition each component mean and scale matrix into observed and target blocks. For the centered observation residual $r_k = y - \mu_{y,k}$, define

$$S_k = \Sigma_{yy,k} + \tau^2 I, \quad \delta_k = r_k^\top S_k^{-1} r_k,$$

where $\tau = 3 \times 10^{-4}$ is a validation-selected model-mismatch floor, not a calibrated camera-noise variance.

The observed log density, omitting the phase-common constant, is

$$\ell_k = \log \pi_k - \frac{1}{2} \log |S_k| - \frac{\nu + 49}{2} \log \left(1 + \frac{\delta_k}{\nu} \right).$$

Posterior responsibilities use a fixed temperature $T = 4$:

$$\gamma_k = \frac{\exp(\ell_k/T)}{\sum_j \exp(\ell_j/T)}.$$

The conditional Student-t location is the same affine form as Gaussian mixture regression [9]:

$$m_k(y) = \mu_{t,k} + \Sigma_{ty,k} S_k^{-1} (y - \mu_{y,k}).$$

The two missing center values are estimated by the posterior mean of component locations,

$$\hat{t} = \sum_k \gamma_k m_k(y).$$

The conditional predictive scale depends on δ_k and is retained for calibration experiments, but it is not required to form the conditional location. All factorizations use Cholesky solves, and responsibility normalization uses log-sum-exp in the reference implementation.

Finally, the measured center component is copied exactly from the input mosaic. Exact reinjection is important: even mathematically normalized posterior weights can change a measured float by a rounding unit if all three channels are averaged uniformly.

4.1 Why Student-t responsibilities

A Gaussian likelihood penalizes a large Mahalanobis distance exponentially. When a patch lies outside the population distribution, tiny covariance or tail differences can produce a nearly one-hot yet incorrect component assignment. A Student-t component replaces this exponential tail with a polynomial one. In the frozen Gaussian experts, changing only the component likelihood to Student-t increases held-out BSDS PSNR from 34.835 to 35.404 dB. Continuing the parameters with true Student-t ECM adds a further 0.114 dB, reaching

35.518 dB. Most of the improvement therefore comes from robust responsibility assignment rather than from altered affine experts.

4.2 Fixed-degree-of-freedom ECM

Training begins from a deterministic full-covariance Gaussian mixture. For a sample z_i and component k , the E step computes the posterior responsibility and the standard latent Student-t scale weight

$$u_{ik} = \frac{\nu + 51}{\nu + \text{delta}_{ik}}.$$

The conditional maximization updates mixture weights from responsibilities and updates component locations and scales using the products $\gamma_{ik}u_{ik}$, with covariance floor 10^{-6} . Degrees of freedom are held fixed. This follows the robust-mixture latent-scale formulation of Peel and McLachlan [8], independently implemented without copying reference source. Validation selected $\nu = 3$ and 30 continuation iterations. Training likelihood rises monotonically; validation is nearly flat from iterations 5 through 30 and falls by only 0.061 dB at iteration 50.

5 Model Reduction and Component Shortlisting

The full model has 64 components in every phase and evaluates a complete 49-D likelihood and predictor for every component. Reduction was performed only after the statistical configuration and safety tests were frozen.

5.1 Genuine K=32 training

Models with 8, 16, and 32 components were independently initialized from their corresponding Gaussian checkpoints and continued with Student-t ECM. They were not formed by slicing the K=64 model. K=32 is the smallest model that retains at least 90% of the validation MSE gain: its dense test result is 35.450 dB, compared with 35.518 dB for K=64. K=16 remains strong but misses the predeclared retention criterion, while K=8 regresses on a bright-star safety subset.

5.2 S9/q8 observable shortlist

Every target still uses the fixed 7 by 7 observed patch. There is no nonlocal patch retrieval. The term selection refers to two deterministic operations:

1. choose one of 18 phase banks by matching the actual CFA pattern;
2. rank mixture components using an observable central marginal.

For each K=32 component, the algorithm evaluates a Student-t marginal on the nine physical samples in the central 3 by 3 region (S9). Stable insertion selection retains the top eight component identifiers, with ties resolved by the lower identifier. Only those eight components receive the full 49-D likelihood and conditional prediction. The observable RGB DC

is still formed from all 49 samples; changing its support would change the statistical model rather than only the search cost.

On untouched test coordinates the shortlist contains the exact dense-K32 top component 90.53% of the time and retains 83.31% mean exact posterior mass. The combined K32/S9/q8 method retains 90.81% of the full K64 test MSE gain while reducing the arithmetic estimate from 163,072 to 23,264 MAC-scale operations per center, a factor of 7.01.

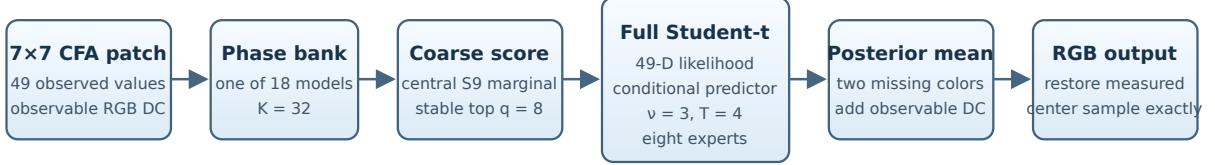


Figure 1: TGMR inference pipeline. The observed patch is fixed; only the phase bank and mixture components are selected.

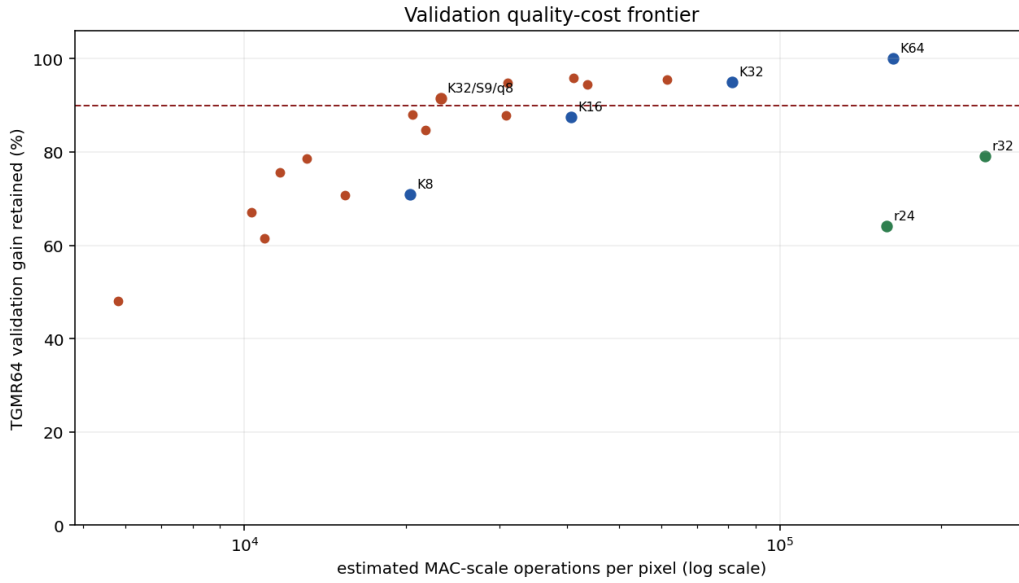


Figure 2: Validation quality-cost frontier used to freeze K32/S9/q8.

6 Exact Native Inference

6.1 Algebraic specialization

The frozen parameters allow exact removal of per-pixel transcendental calls. For S9, $\nu = 3$ gives the component score

$$C_k - 6 \log(1 + \delta_k/3),$$

where C_k contains the mixture weight and determinant terms. Ordering can be computed from the positive quantity

$$\frac{\exp((C_k - C_{\max})/6)}{1 + \delta_k/3}.$$

The phase-common subtraction prevents overflow and does not affect ranking. For the full 49-D likelihood with $T = 4$, the posterior weight is proportional to

$$\exp((C_k - C_{\max})/4) (1 + \delta_k/3)^{-6.5} = \frac{B_k}{s^6 \sqrt{s}}, \quad s = 1 + \delta_k/3.$$

The factors involving only model coefficients are precomputed. This is an algebraic specialization, not a likelihood approximation. Across 1,152 phase probes, the specialized scalar and AVX2 paths produce zero shortlist mismatches relative to the logarithmic reference.

6.2 SIMD organization

The S9 model is stored coefficient-major in four array-of-structures-of-arrays groups. Each AVX2 vector solves eight independent component systems for one pixel. Stable scalar insertion selects component identifiers.

The more important optimization is applied to the full stage. Shortlisted requests are bucketed locally by component. An AVX2 vector then processes eight pixels that share one component, broadcasting that component's 49-D Cholesky coefficients and two conditional gain rows. This avoids gathers across unrelated experts. Incomplete vector groups use a scalar tail. The implementation does not use `-ffast-math`, approximate reciprocal square root, or an approximate statistical model.

6.3 OpenMP and bounded streaming

The engine flattens (phase, chunk) work and schedules it statically in one OpenMP loop. A 128 by 128 tile is a scheduling unit, not a reconstruction boundary: every 7 by 7 patch is read from the complete input mosaic. Only the true image boundary is reflected, without repeating the edge sample. Tile coordinate lists are grouped by phase, while 49-value observations are materialized only for the current 512-pixel chunk.

The worker scratch is 419,840 bytes. At 18 workers, worker scratch is about 7.2 MiB; the model is approximately 6 MiB and the prepared AoSoA cache about 210 KiB. No pixels-by-49 full-frame feature matrix is allocated.

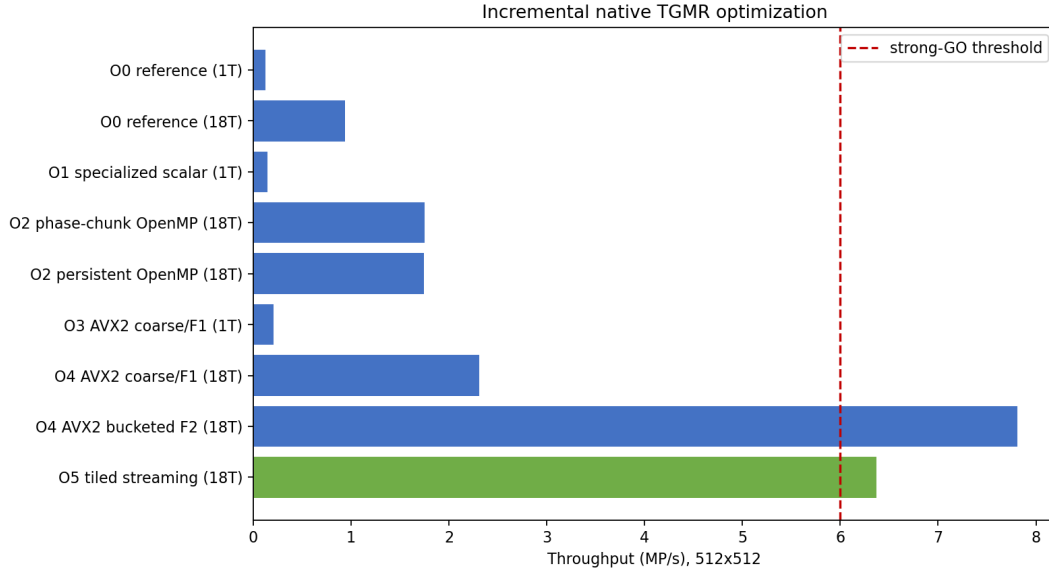


Figure 3: Incremental native optimization. The largest gain comes from same-component full-likelihood SIMD.

7 Training and Evaluation Protocol

7.1 Population data

Natural-image training uses a mirror of the BSDS500 segmentation corpus at revision a04b7c6c3a9f0ace74bf205c72a43d32e1c72722 [12]. Stored sRGB JPEG images are decoded to linear light before patch extraction and synthetic mosaicking. The dataset’s research and educational terms apply independently of this paper.

Split	Sources	Sampling per source	Patches
Training	200 official training images	512 deterministic random 7 by 7 patches	102,400 per phase
Validation	20 frozen validation images	dense 24 by 24 center grid	11,520
Test	20 untouched test images	dense 24 by 24 center grid	11,520

Every training RGB patch is presented under each of the 18 phase contracts, so each phase model receives 102,400 examples. Training patch positions are selected without replacement using deterministic SHA-256-derived seeds. The Gaussian initialization uses deterministic full-covariance EM implemented with scikit-learn [13]; four training threads and one evaluation thread are part of the reproduction contract because BLAS reduction order can change final float bits.

Model and hyperparameter selection use only training and validation. The untouched test,

external images, star fields, and analytical scenes are opened only after the configuration is frozen.

7.2 External and analytical controls

The external chromatic suite contains three fixed 12 by 12 grids from each of Chelsea, Coffee, IHC, Motorcycle, Retina, and Rocket, totaling 2,592 target pixels. Hubble and an independent NASA Hydra image contribute uniform and bright-target star samples. Six analytical scenes - two gradients, a red/gray edge, periodic chromatic texture, a tiny star, and a saturated point - are evaluated under all 18 X-Trans phases.

The primary quality metric pools squared error over the two predicted center channels at fixed coordinates:

$$\text{PSNR} = 10 \log_{10} \frac{1}{\text{MSE}},$$

for normalized linear RGB. Reported headline PSNR is therefore a center-pixel conditional-reconstruction metric, not full-image rendered PSNR. We also report p99 absolute channel error, per-source deltas, phase spread, exact native-sample restoration, and frozen safety cases.

7.3 Comparisons

Markesteijn and corrected-final MLRI are classical X-Trans implementations. GMAX is a same-population global joint spatial-chromatic affine LMMSE bank. FULL-GMM is a 16-component, 147-dimensional full-RGB patch mixture. Gaussian phase GMR is the direct 51-dimensional predecessor of TGMR. All center-pixel methods are evaluated on identical coordinates.

8 Results

8.1 Main held-out result

Method	Pooled PSNR (dB)	p99 absolute error
Markesteijn	30.867	0.137975
corrected-final MLRI	30.985	0.134802
GMAX LMMSE	32.054	0.122019
FULL-GMM	33.531	0.099805
Gaussian phase GMR	34.835	0.084625
Gaussian experts, Student-t responsibilities	35.404	0.078690
trained K64 Student-t GMR	35.518	0.077291
reduced K32/S9/q8 TGMR	35.056	0.081494

The reduced method is 3.002 dB above GMAX and 4.189 dB above Markesteijn on the fixed-coordinate test. It retains 90.81% of the K64 MSE improvement over GMAX. Relative to Gaussian GMR, the full K64 Student-t model reduces p99 error by 8.67%. The K64 model wins 18 of 20 test sources over Gaussian GMR; the two losses are 0.296 and 0.043 dB.

The progression also isolates why the method succeeds. Task reduction from FULL-GMM to Gaussian phase GMR adds 1.304 dB. Reinterpreting only the responsibilities with Student-t tails adds 0.569 dB. Robust Student-t fitting adds 0.114 dB. The final reduction gives back 0.462 dB for a 7.01-fold arithmetic reduction.

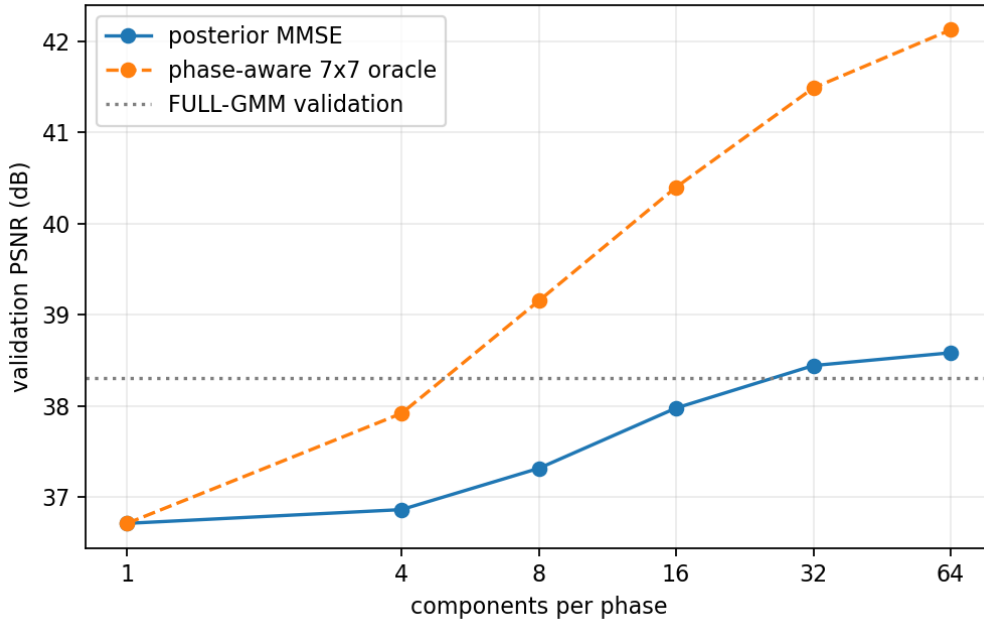


Figure 4: Validation quality as Gaussian components are added to the phase-conditioned task regressor.

8.2 Heavy-tail robustness

With fixed Gaussian experts, validation PSNR has a broad maximum for $\nu = 3$ through 10 rather than a fragile single optimum. At $\nu = 5$, mean maximum responsibility falls from 0.653 for the Gaussian likelihood to 0.494, and effective component count rises from 3.55 to 6.59. After Student-t ECM, validation selects $\nu = 3$.

The critical external failure of Gaussian GMR was motorcycle-left-0, at -0.853 dB relative to GMAX. Student-t responsibilities alone move it to +2.435 dB, and trained TGMR reaches +2.811 dB. This crop was not used for selection. For the final reduced model, all 18 external crops remain within the frozen -0.5 dB safety bound; the worst is coffee-1 at -0.314 dB.

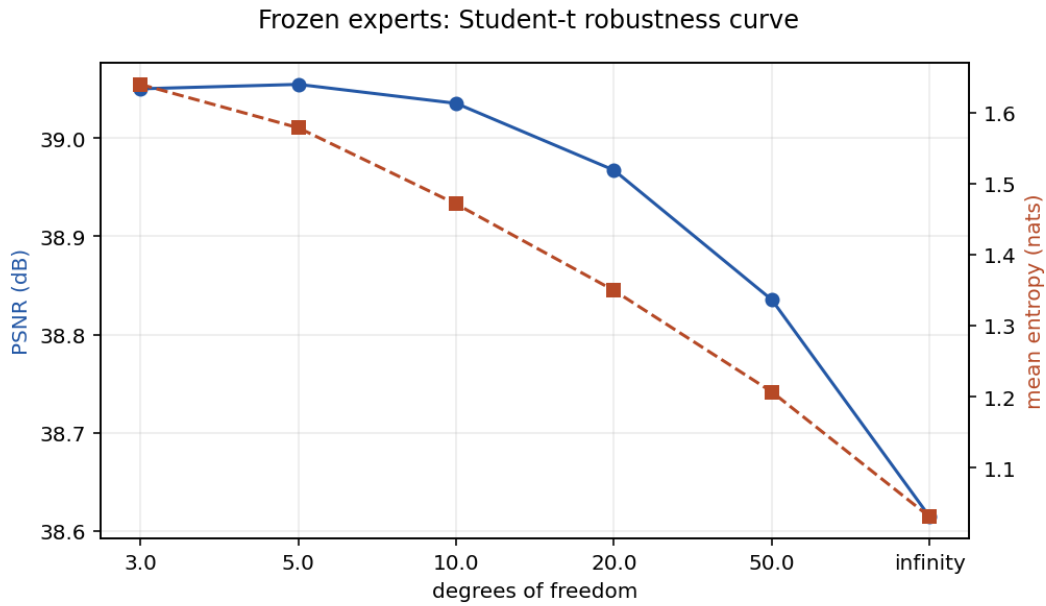


Figure 5: Validation PSNR and responsibility behavior as the Student-t tail changes.

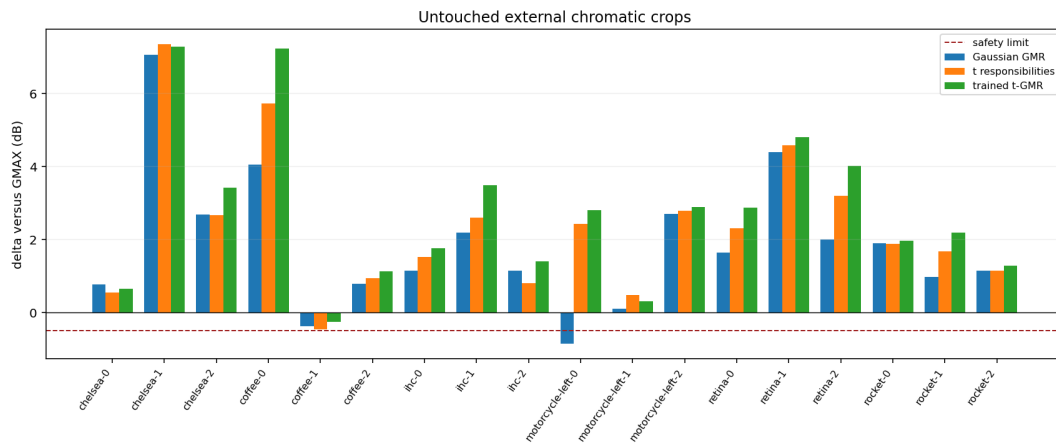


Figure 6: External chromatic crop deltas of the full trained Student-t model relative to GMAX.

8.3 Sparse and phase-sensitive structures

Control	GMAX (dB)	K64 TGMR (dB)	K32/S9/q8 (dB)	Reduced delta vs GMAX
Hubble bright	15.435	15.707	15.921	+0.486
Hydra bright	32.784	40.165	36.447	+3.663
gray gradient	69.907	76.609	75.398	+5.491
chromatic gradient	59.401	70.101	69.812	+10.411
red/gray edge	19.537	27.890	26.577	+7.040

Control	GMAX (dB)	K64 TGMR (dB)	K32/S9/q8 (dB)	Reduced delta vs GMAX
periodic chromatic	16.175	18.574	18.454	+2.279
tiny star	8.539	8.224	8.267	-0.271
saturated point	3.807	3.984	4.018	+0.211

The star-field controls prevent a natural-image average from hiding sparse failures. Reduced TGMR improves both bright subsets over GMAX, but neither mixture model surpasses Markestijn on Hubble bright in the full study. The tiny-star analytical scene remains a visible limitation: TGMR is stable and inside the predeclared -2 dB bound, but 0.271 dB below GMAX. A heavy-tailed population prior does not recover information absent from the CFA, and it can still prefer a statistically common smooth explanation for a one-pixel structure.

Phase behavior is mixed. Reduction decreases the K64 phase-PSNR range on five of six analytical scenes, but the absolute range remains large for the two gradients and periodic texture. Phase-conditioned training handles each local sampling pattern correctly; it does not make the CFA information content phase invariant.

8.4 Numerical parity

Float32 model and inference differ from the float64 selected result by 2.59×10^{-8} RMS and 3.18×10^{-7} maximum over the frozen evaluation. The final AVX2/FMA executor differs from the frozen float32 reference by 4.29×10^{-8} RMS and 4.77×10^{-7} maximum over 15,564 evaluation samples. No non-finite value occurs, shortlist identifiers match on all audit probes, output hashes are identical at 1, 12, 18, and 24 threads, and measured center samples are exact.

8.5 Runtime and memory

All native timings use an AMD Ryzen 9 5900X and GCC 13.3. Medians follow one warm-up. The final standalone streaming executor uses 18 OpenMP workers, AVX2/FMA, 128 by 128 tiles, and 512-pixel chunks.

Geometry	Pixels	Median (s)	Throughput (MP/s)	Peak RSS (MiB)
2048 by 2048	4.19 M	0.603	6.957	85
4000 by 3000	12.00 M	1.749	6.862	204
6000 by 4000	24.00 M	3.487	6.883	387
7738 by 5164	39.96 M	5.833	6.850	631

Throughput is nearly flat from 4 to 40 megapixels. The large-run memory is explained by scalar input and RGB output planes plus bounded scratch, not by a full-frame observation matrix. Scaling peaks at 18 hardware threads; using all 24 SMT threads reduces throughput.

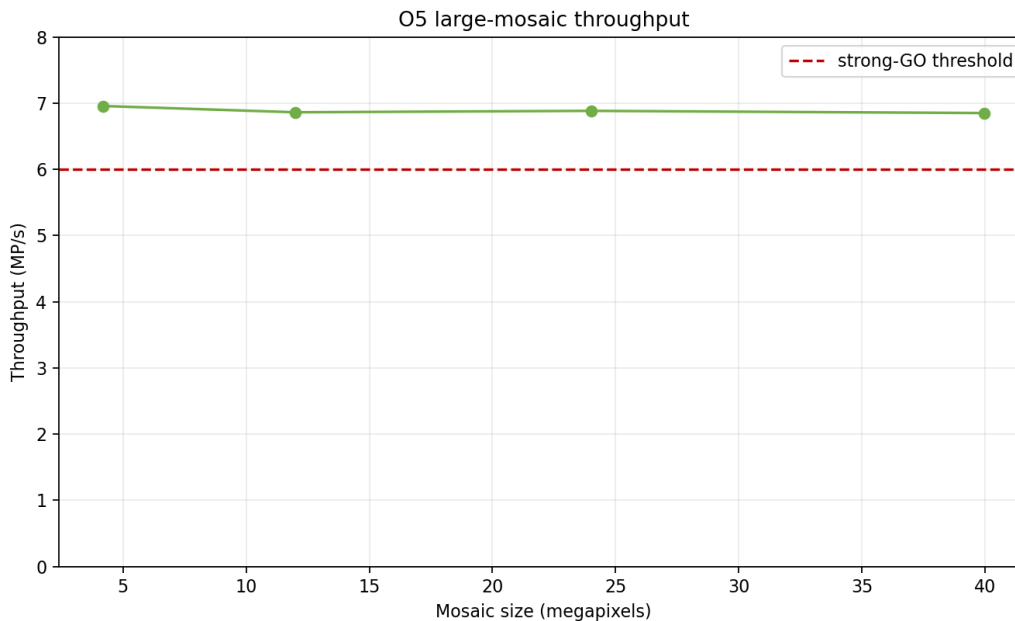


Figure 7: Standalone streaming throughput remains approximately constant through 40 megapixels.

9 RawTherapee Integration

The experimental method is registered as `tgmr`, displayed as **TGMR (experimental)** [5]. It loads one external binary artifact of 6,073,164 bytes with SHA-256 6279b6a593ef4b595b1aff246182682b60c7eea701373ee2eb9f80cfcb50485c. The loader authenticates the complete file, then validates dimensions, indices, phase patterns, center/target channels, S9 support, finite values, positive Cholesky diagonals, and canonical end of file. Failed loads are not cached.

RawTherapee supplies a black-subtracted, preprocessing-balanced camera-sensor mosaic in its nominal 0 to 65535 float scale. TGMR divides by 65535 before inference and multiplies its predictions afterward. It applies no additional white balance, color matrix, clipping, false-color suppression, sharpening, or denoising. The measured channel is copied from the original engine input, not round-tripped through the normalization.

The engine matches the camera’s actual 6 by 6 CFA against all authenticated phase patterns. Any load, CFA, allocation, non-finite, or inference failure discards partial output and reruns Markestijn over the complete RGB planes. The fallback is loud and carries a structured error code.

On a deterministic 39.96-megapixel mosaic, the integrated kernel reaches 5.248 MP/s, 79.2% of a contemporaneous standalone run. Four 40-megapixel Fujifilm X-T50 RAFs spend 7.34 to 7.47 s in the kernel and 10.78 to 11.28 s in the complete minimal CLI pipeline. For DSCF0771, Markestijn takes 3.78 s and TGMR 11.28 s, about a factor of three. Peak RSS is effectively identical near 2 GiB for both complete exports.

Visual inspection of these four RAFs found plausible color and no obvious tile seam or global cast. On one metallic earring, TGMR reduces visible magenta/cyan segmentation relative to Markestijn; fine foliage and cobblestones are somewhat smoother. These are qualitative

observations from one camera model without ground truth. No identifiable real-image figure is included in this paper, and the observation is not treated as a general quality claim.

10 Discussion

10.1 What the mixture learns

The large gap between $K=1$ and the mixture shows that one global covariance does not capture all useful joint spatial-chromatic behavior. A phase-local component bank supplies edge, texture, and chromatic alternatives, while posterior averaging avoids brittle hard switching. The component oracle still reaches 38.684 dB with 7 by 7 selection and 45.239 dB per pixel for K64, so the learned experts contain more capacity than the observable posterior can recover. That gap is not an invitation to train an external selector: component identifiers have phase-local meaning, and earlier hard-selection experiments were unsafe.

Student-t tails address a narrower failure. They reduce the influence of a large Mahalanobis residual on component choice, making atypical observations less likely to produce a one-hot Gaussian assignment. The result is unusual in that robustness and average quality improve together. The remaining coffee-1 and tiny-star regressions show that heavy tails do not eliminate model mismatch or CFA ambiguity.

10.2 Why task reduction matters

FULL-GMM models all 147 RGB variables even though inference needs only two center colors. Density components that explain neighboring unobserved RGB do not necessarily minimize target error. The 51-dimensional task model removes those nuisance variables and trains one population for each actual observation operator. This increases validation stability, makes $K=1$ exactly comparable to same-support LMMSE, and improves external generalization before Student-t robustness is introduced.

10.3 Cost-quality tradeoff

The K32/S9/q8 candidate is a deliberately frozen knee rather than the maximum quality point. Genuine K32 training is superior to retaining the most common 32 K64 components. Low-rank covariance approximation is not useful: even rank 24 retains only 64.14% of the validation gain while costing nearly as much as dense inference. Observable S9 shortlisting is less accurate than an S25 marginal but substantially cheaper and remains above the 90% validation rule.

The successful native implementation changes the computational organization, not the model. Same-component bucketing creates the SIMD axis missing from a naive per-pixel triangular solver. Bounded streaming then removes hundreds of megabytes of observation materialization for only a small throughput cost.

11 Limitations and Reproducibility Boundary

The principal limitations are:

- **Evaluation unit.** Headline PSNR is measured at fixed center coordinates, not over every pixel of a reconstructed full image. The engine integration establishes numerical and qualitative behavior but does not provide ground-truth full-RAF PSNR.
- **Training domain.** BSDS JPEGs are linearized and remosaicked synthetically. Real RawTherapee input is camera-native sensor RGB after black subtraction and preprocessing balance. Transfer across sensor generations, illuminants, noise levels, and color matrices is not established.
- **Sparse null-space detail.** The tiny-star control remains below GMAX, and Hubble bright remains below Markestijn in the full comparison. Statistical priors cannot guarantee recovery of unobserved exceptional colors.
- **Phase spread.** Correct phase modeling does not remove phase-dependent information loss. Several analytical PSNR ranges remain large.
- **Posterior calibration.** Predictive-risk correlation improves under Student-t fitting, but confidence is not reliable enough to define a safe automatic fallback gate.
- **Model rights.** The fitted model derives from BSDS training material and remains external while redistribution terms are unresolved. The CC BY 4.0 license of this paper does not license the model or dataset.
- **Hardware scope.** Optimized timings describe one Zen 3 CPU. A scalar fallback exists, but equivalent throughput is not claimed on other architectures.

Reproduction is anchored to tracked canonical JSON, deterministic patch coordinates, fixed thread counts, and logical model hashes. The final native model artifact is intentionally not committed. The paper’s verification script checks the headline numbers and hashes before rendering.

12 Conclusion

Phase-conditioned Student-t mixture regression provides a practical classical statistical alternative for X-Trans demosaicing. The method conditions directly on the 49 physical samples of a 7 by 7 neighborhood, predicts only the two missing center channels, and restores the measured center sample exactly. Heavy-tailed responsibilities correct a demonstrated Gaussian-generalization failure while improving held-out quality. A genuinely trained K32 model with S9/q8 observable shortlisting retains 90.81% of the full Student-t gain and reaches 35.056 dB on untouched BSDS coordinates, 3.002 dB above global LMMSE.

The implementation result is equally important: algebraic specialization, component-bucketed AVX2/FMA, phase/chunk OpenMP, and bounded streaming turn a 0.98 MP/s research kernel into a 6.85 MP/s standalone executor without changing the frozen statistical contract. RawTherapee integration retains 79.2% of that throughput and provides complete authenticated fallback.

The method should remain experimental. A broader camera-linear corpus, cross-camera ground truth, and resolved model licensing are necessary before routine distribution to users. Within those limits, the study demonstrates that a robust, task-reduced mixture of joint spatial-chromatic regressors can capture substantial nonlinear population information without a neural runtime or a brittle external content classifier.

13 Artifact and Ethics Statement

The manuscript source, original diagram, and rendered paper are released under CC BY 4.0. RawTherapee remains under its project license. BSDS500, cited papers, source images, and the external learned model retain their own terms. No identifiable comparison photograph is reproduced.

AI-assisted tools were used for code generation, experiment orchestration, data checking, bibliography preparation, and drafting. The named author is responsible for verifying the mathematical statements, numerical claims, citations, interpretation, and final text.

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